

Name _____

Tie Breaker: Points scored on Stated and Geometry Problems

+ _____ + _____ + _____

5x (Last Problem Attempted) + _____ + _____ + _____

7x (Number Incorrect) - _____ - _____ - _____

2x (Number Incorrect SDs) - _____ - _____ - _____

TOTAL SCORE _____

Grader Initials _____

UIL Calculator Applications

Test 26F (District)

DO NOT OPEN THE TEST UNTIL INSTRUCTED TO BEGIN

I. Calculator Applications rules and scoring—See UIL Constitution

II. How to write the answers

A. For all problems except stated problems as noted below—write three significant digits.

1. Examples (* means correct but not recommended)

Correct: 12.3, 123, 123.*, 1.23x10*, 1.23x10⁰*, 1.23x10¹, 1.23x10⁰¹, .0190, 0.0190, 1.90x10⁻²

Incorrect: 12.30, 123.0, 1.23(10)², 1.23*10², 1.230x10², 1.23*10², 0.19, 1.9x10⁻², 19.0x10⁻³, 1.90E-02

2. Plus or minus one digit error in the third significant digit is permitted.

3. An answer with trailing zeroes may be written in fixed notation without specifying which zeroes are significant. For example, a correct answer is 500. However, it must be written as 5.00x10² in scientific notation to be correct.

B. For stated problems

1. Except for integer, dollar sign, and significant digit problems, as detailed below, answers to stated problems should be written with three significant digits.

2. Integer problems are indicated by (integer) in the answer blank. Integer problems answers must be exact, no plus or minus one digit, no decimal point or scientific notation.

3. Dollar sign (\$) problems should be answered to the exact cent, but plus or minus one cent error is permitted. Answers must be in fixed notation. The decimal point and cents are required for exact-dollar answers.

4. Significant digit problems are indicated by underlined numbers and by (SD) in the answer blank. See the UIL Constitution and Contest Manual for details.

III. Some symbols used on the test

A. Angle measure: rad means radians; deg means degrees.

B. Inverse trigonometric functions: arcsin for inverse sine, etc.

C. Special numbers: π for 3.14159 ...; e for 2.71828 ...

D. Logarithms: Log means common (base 10); Ln means natural (base e); exp(u) means e^u.

VOLUME CONTROL R15A 40K TAP @ 10K

SPEAKER DISCONNECTS

C19 200 P3 +14.4V

INSTRUMENT LAMPS

.312-.320 DIA 6 HOLES EQ SP 1/4" DIA

2.27 2.23

4.02 3.98

COLORADO RIVER

PLAN

26F-1. $0.488 + 0.0811 - 4.13$ ----- 1= _____

26F-2. $(-6.98 + 6.6) \times (\pi) - 3.93$ ----- 2= _____

26F-3. $(-0.431 + 0.944 - 0.478)/(-0.434) + 0.0259$ ----- 3= _____

26F-4. $\{(67.8)(0.202 + 1.1 - 0.399)(-42.5)\} + 1720$ ----- 4= _____

26F-5. $\frac{(-0.00809 - 0.00685)(0.0494)}{\{(0.0877)/(0.0808)\}} - (-0.00146 - 7.53 \times 10^{-4})$ ----- 5= _____

26F-6. What is 76.4 raised to the power π ? ----- 6= _____

26F-7. What is the product of positive $\sqrt{0.0599}$ and 0.983? ----- 7= _____

26F-8. What is the remainder of 95.6 divided by 1.55? ----- 8= _____

26F-9.

SQUARE

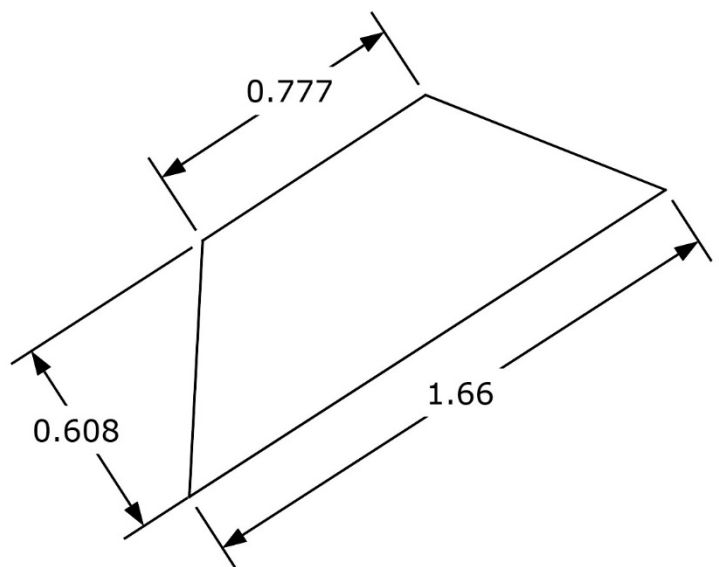


Perimeter = 115

26F-9 = _____

26F-10.

ISOSCELES TRAPEZOID



Area = ?

26F-10 = _____

26F-11. $\frac{(71.4)(21.3) - (-8.11)(-82.1) + 210}{5430 + (88.4)(55.3)}$ ----- 11=_____

26F-12. $\frac{\{-0.00844 + (0.591)(0.187)(-0.0862)\}}{(0.335 + 0.345)(-0.701)(1.8 + 0.827)}$ ----- 12=_____

26F-13. $\frac{(42.2)(937 - 836)\{5900 - (48.4)(34.1)\}}{(78.3 + 62.9)(-14.5 - 17.1)}$ ----- 13=_____

26F-14. $\frac{(9780 + 9460 - 6680)(0.00446 + 0.00655 - 0.0032)}{(-0.224 - 0.0544)(9.37)(0.857 - 0.538)}$ ----- 14=_____

26F-15. $\frac{(19800 + 3420 - 7000)(0.804 - 0.383 - 1.15)}{(198)(-175)(557)(7.32 + 7.27 + 26.7)}$ ----- 15=_____

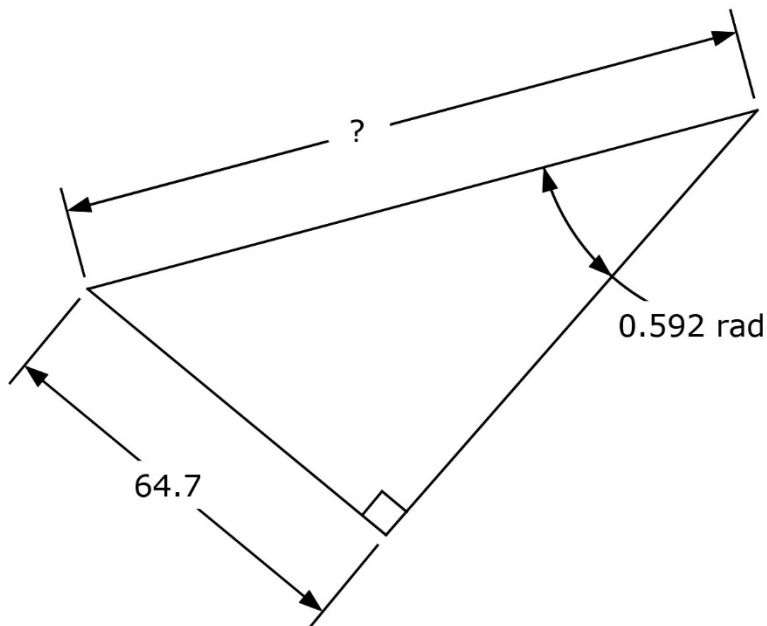
26F-16. Daniel wants to offer 335 hot dogs for his family reunion. If the buns come eight to a package, how many packages should he buy? -----16=_____ integer

26F-17. Human hair grows at an average rate of 0.5 in/month. If a person could grow a single strand of hair for their entire 86-year life, how long would the hair be? -----17=_____ ft

26F-18. A caterer charges \$85 for a gallon of coffee. If a cup is 6 oz, what is the average cost for a cup of coffee? -----18=\$_____

26F-19.

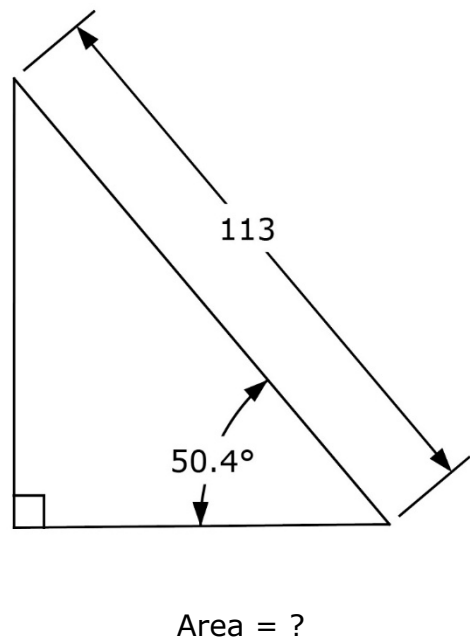
RIGHT TRIANGLE



26F-19 = _____

26F-20.

RIGHT TRIANGLE



26F-20 = _____

26F-21. $\frac{1}{8.64 + 21.6} + \frac{1}{3.74 - 8.31} + \frac{1}{(22.3)}$ ----- 21=_____

26F-22. $\left[\frac{(0.377)(0.691)}{-5.85} + 0.0285 \right]^2 + \sqrt{2.83 \times 10^{-8}}$ ----- 22=_____

26F-23. $\left[\frac{0.516 + 0.365 + \sqrt{0.209/0.93}}{-0.0738 + 0.0115} \right]^2$ ----- 23=_____

26F-24. $(-0.112)(\pi) + \sqrt{(0.283)/(8.06)} + [(0.458)(1.04)]^2$ ----- 24=_____

26F-25. $(-5.82)(-0.147)\sqrt{(-0.732)^2/0.911} + 1/\sqrt{1.37 + \pi}$ ----- 25=_____

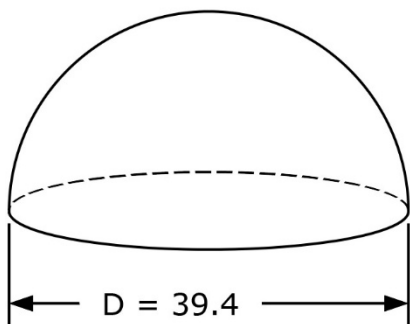
26F-26. A sloth averages 2.7 in/s normally, but it can move at 0.17 mph when threatened. What is the percent change in sloth speed when threatened? -----26=_____ %

26F-27. A car lane is 12 ft wide, and a bicycle lane is 5 ft wide. It costs \$980,000 to repave a 3500 ft long stretch of a 4-lane road. How much would it cost to repave 2 mi of a 5 lane road with two bicycle paths? -----27=\$_____

26F-28. Sound travels at 1125 ft/s. Shirley yells at the top of a canyon, and she hears the echo after 2.8 s. What is the distance across the canyon? -----28=_____ ft

26F-29.

HEMISPHERE

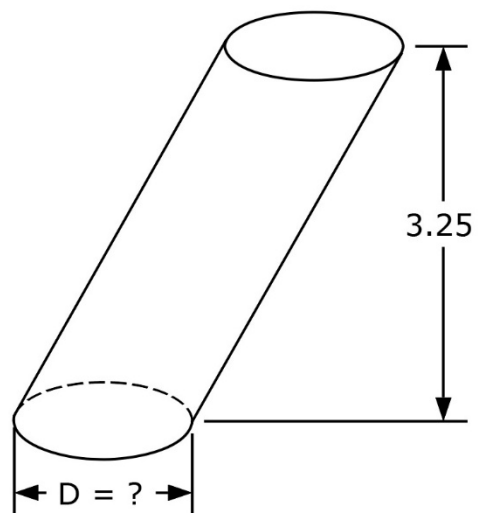


Total Surface Area = ?

26F-29 = _____

26F-30.

SLANT CIRCULAR CYLINDER



Volume = 6.13

26F-30 = _____

$$26F-31. \frac{(-832 + 855)^2}{\sqrt{20.2 - 8.55}} + \frac{4260}{\sqrt{525 + 636}} \text{ ----- } 31 = \underline{\hspace{2cm}}$$

$$26F-32. \sqrt{\frac{1/(859 - 437)}{(142)(1.32 + 0.597)^2}} + (47.7)^2(7.68 \times 10^{-7}) \text{ ----- } 32 = \underline{\hspace{2cm}}$$

$$26F-33. \frac{[0.0204/(0.685 + 0.818) + 1/(32.6)]^{1/2}}{(23.3 + 128)^2 \times \sqrt{55.6 - (-26.2)}} \text{ ----- } 33 = \underline{\hspace{2cm}}$$

$$26F-34. \frac{\sqrt{(74300)/\{(3070)/\sqrt{84600}\}}}{79.9 + (0.563)(1.92)} + \{\pi + 1.2\}^{1/2} \text{ ----- } 34 = \underline{\hspace{2cm}}$$

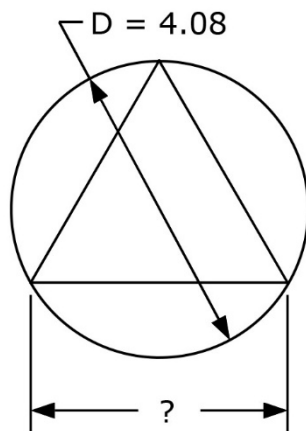
$$26F-35. \frac{\left[\frac{4.48 \times 10^{-4}}{927}\right]^2 + \sqrt{\frac{(0.931)(0.917)}{(1.63 \times 10^{25})}} + (2.09 \times 10^{-12})}{0.326 + \sqrt{(-0.696)(-0.532)}} \text{ ----- } 35 = \underline{\hspace{2cm}}$$

26F-36. A city doubles in population every 46 years. After how many years will the population triple? ----- 36 = yr

26F-37. Saran wrap, 1/2000 in thick, is a thin plastic sheet spooled onto a 1 in diameter roll. It is sold in 100 ft² packages that are 18 in wide. What is the outer diameter of a spool of Saran wrap? ----- 37 = in

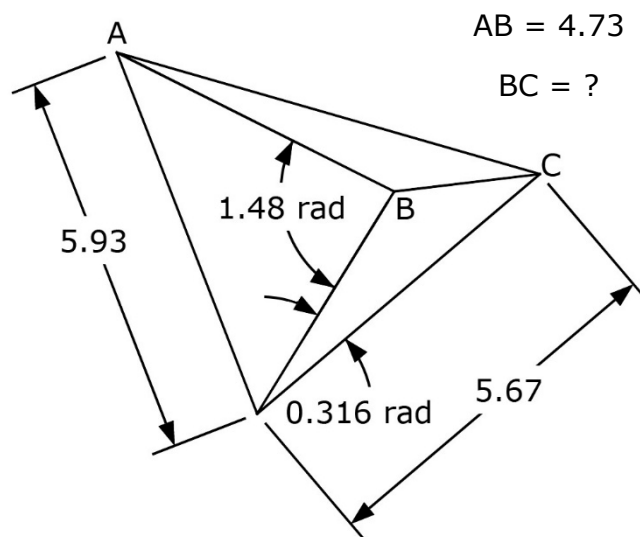
26F-38. China produced 207.5×10^6 metric tons of rice in 2024. A metric ton is 1000 kg. If all this rice were placed in a cube of side dimension A, what is A? A grain of rice weighs 0.0289 g, and there are 190 g of rice in a cup. ----- 38 = ft(SD)

26F-39.
CIRCUMSCRIBED EQUILATERAL TRIANGLE



26F-39 =

26F-40.
SCALENE TRIANGLES



26F-40 =

26F-41. $\frac{10^{-(4.5 - 7.36)}}{5.87 + 2.79}$ ----- 41=_____

26F-42. $\frac{(-3.45)}{(-9.92)} [1 - e^{-(0.218)(0.454)}]$ ----- 42=_____

26F-43. $(7700 - 21800) \ln\{(-6360)(-5090)\}$ ----- 43=_____

26F-44. $(-6420 + 11600)^{-(0.931 + 0.442)}$ ----- 44=_____

26F-45.(deg) $\sin \left[90^\circ \times \frac{(-0.00854)}{(0.0126)} \right] + \cos \{94^\circ - 23.1^\circ\}$ ----- 45=_____

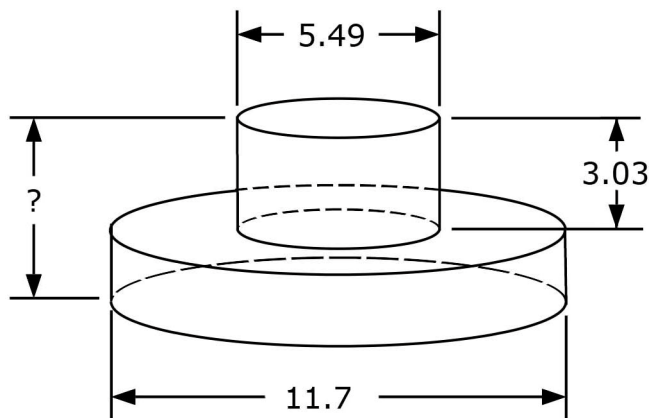
26F-46. Zest is the thin surface skin of a fruit. If a recipe calls for the zest of 22 Persian limes, 2.5 in long, how many key limes would be needed to provide the same amount of zest? A key lime is 1.5 in long. -----46=_____ limes (integer)

26F-47. The wingspan (in) and weights (lbs) of various birds are: titmouse (6.1, 0.03), robin (14.2,0.17), blue jay (17, 0.29), and woodpecker (26, 0.66). What is the wingspan of a bald eagle weighing 13 lbs? -----47=_____ in

26F-48. For what negative value of x does $4^x = x^4$? -----48=_____

26F-49.

CYLINDERS

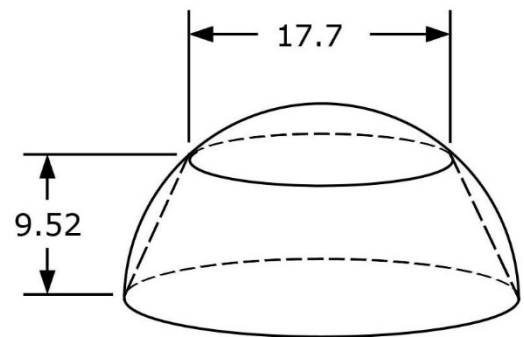


Large
Cylinder= 3 $\left[\begin{array}{c} \text{Small} \\ \text{Cylinder} \\ \text{Volume} \end{array} \right]$

26F-49 = _____

26F-50.

FRUSTUM INSIDE HEMISPHERE



Volume(Frustum) = ?

26F-50 = _____

26F-51. $\frac{10^{(0.55)} \times 10^{-(0.311)} + 0.32}{10^{(4.82 + 0.367)}} \text{ ----- } 51 = \underline{\hspace{2cm}}$

26F-52. $\frac{1 + e^{\{0.862 + (0.125)(1.11)\}}}{(-8.21 \times 10^{-5})(\pi - e^{(-0.95)})} \text{ ----- } 52 = \underline{\hspace{2cm}}$

26F-53. $\frac{\text{Ln}(7840 + 35700)}{5130} + \frac{\text{Ln}(8120)}{6790 - 1700} \text{ ----- } 53 = \underline{\hspace{2cm}}$

26F-54. $\frac{(0.703 + 0.788)^{-0.866}}{(0.407)^{-(0.822 + 0.439)}} \text{ ----- } 54 = \underline{\hspace{2cm}}$

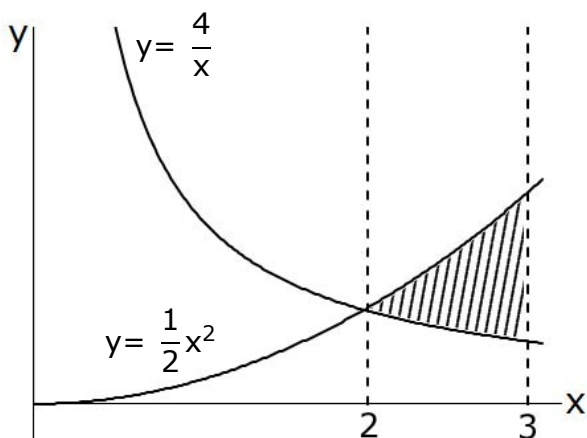
26F-55.(rad) $\arctan \left[\frac{(2920)(0.226)}{(7.62)(36.3)} \right] + (0.471)(1.65) \text{ ----- } 55 = \underline{\hspace{2cm}}$

26F-56. (rad) Calculate the area under the curve $y = 6 - 3x \sin(3x)$ between $x = 0$ and $x = 2$. ----- 56 =

26F-57. Farmer Peter has 150 ft of fencing to outline an enclosure. The enclosure shape is a semicircle attached to a rectangle, . What is the diameter of the semicircle that maximizes the enclosure area? ----- 57 = ft

26F-58. What is M_2 if $\mathbf{M} = \mathbf{NO}$, $\mathbf{N} = \begin{bmatrix} 69 & 13 \\ 13 & 54 \end{bmatrix}$ and $\mathbf{O} = \begin{bmatrix} 25 \\ -31 \end{bmatrix}$? ----- 58 =

26F-59.

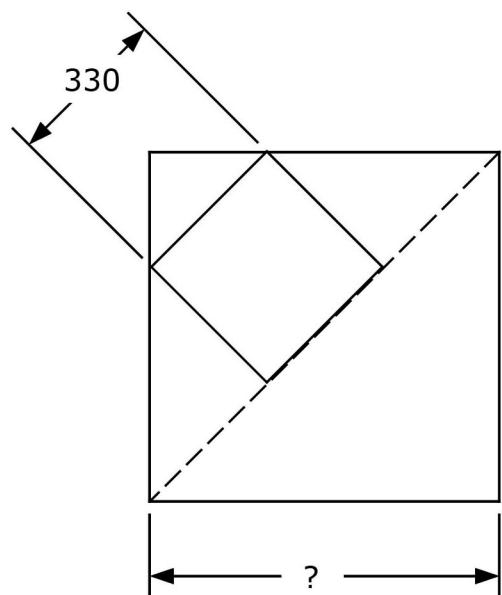
SOLID OF REVOLUTION ($x = 2$)

Volume = ?

26F-59 =

26F-60.

SQUARES

26F-60 =

26F-61. Mike and Kathleen proof documents for a publisher. Mike starts proofing Shakespeare's *Romeo and Juliet* at 8 AM, reading 75 words/min.

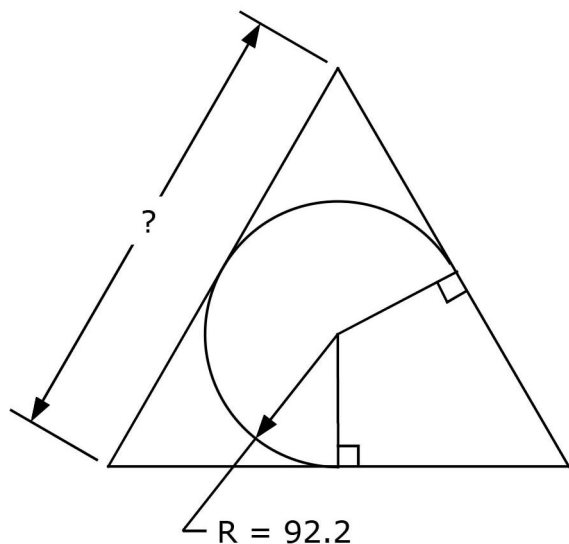
Two hours later, Kathleen joins him, proofing different sections at

88 words/min. They finish at 11:43 AM. How many words are in the play? ---61=_____ words

26F-62. What is Avogadro's Number, 6.02×10^{23} g/mol, raised to the power 863? -----62=_____

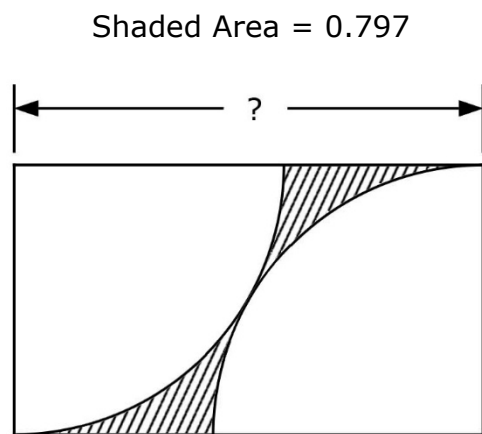
26F-63. Jean fires a projectile at 84 mph and at an angle relative to the ground of 37° . It falls 37 ft short of its target. What should the new velocity be to hit the target? -----63=_____ mph

26F-64.
EQUILATERAL TRIANGLE AND SECTOR



26F-64 = _____

26F-65.
RECTANGLE AND QUARTER CIRCLES



26F-65 = _____

26F-66. $\text{Log}[(1.14)^{-0.226}] + (5.71)\text{Log}[(1.14)^{(1.4)}]$ ----- 66=_____

26F-67. $(88.4)10^{\text{Log}[(5.88)(0.638)]} + \{(1.31 \times 10^5)(0.926)\}^{1/2}$ ----- 67=_____

26F-68. (rad) $(8.67) \left[\frac{\cos(-4.47)}{(-4.47)} + \frac{\cos(4.2)}{(4.2)} \right]$ ----- 68=_____

26F-69. $(0.89) - \frac{(0.89)^2}{2} + \frac{(0.89)^3}{3} - \frac{(0.89)^4}{4}$ ----- 69=_____

26F-70. $\frac{1}{\sqrt{(63.3)^2 - (1310)}} \text{Ln} \left\{ \frac{(60.6) - \sqrt{(63.3)^2 - (1310)}}{(60.6) + \sqrt{(63.3)^2 - (1310)}} \right\}$ ----- 70=_____

26F-1	= -3.56 = -3.56×10^0	26F-11	= 0.103 = 1.03×10^{-1}	26F-21	= -0.141 = -1.41×10^{-1}
26F-2	= -5.12 = -5.12×10^0	26F-12	= 0.0143 = 1.43×10^{-2}	26F-22	= 0.000425 = 4.25×10^{-4}
26F-3	= -0.0547 = -5.47×10^{-2}	26F-13	= -4060 = -4.06×10^3	26F-23	= 473 = 4.73×10^2
26F-4	= -882 = -8.82×10^2	26F-14	= -118 = -1.18×10^2	26F-24	= 0.0624 = 6.24×10^{-2}
26F-5	= 0.00153 = 1.53×10^{-3}	26F-15	= 1.48×10^{-5}	26F-25	= 1.13 = 1.13×10^0
26F-6	= 824000 = 8.24×10^5	26F-16	= 42 integer	26F-26	= 10.8 = 1.08×10^1
26F-7	= 0.241 = 2.41×10^{-1}	26F-17	= 43.0 = 4.30×10^1	26F-27	= \$4,312,000.00
26F-8	= 1.05 = 1.05×10^0	26F-18	= \$3.98	26F-28	= 1580 = 1.58×10^3
26F-9	= 28.8 = 2.88×10^1	26F-19	= 116 = 1.16×10^2	26F-29	= 3660 = 3.66×10^3
26F-10	= 0.741 = 7.41×10^{-1}	26F-20	= 3140 = 3.14×10^3	26F-30	= 1.55 = 1.55×10^0

26F-31	= 280 = 2.80×10^2	26F-41	= 83.7 = 8.37×10^1	26F-51	= 1.34×10^{-5}	26F-61	= 25800 = 2.58×10^4
26F-32	= 0.00388 = 3.88×10^{-3}	26F-42	= 0.0328 = 3.28×10^{-2}	26F-52	= -16400 = -1.64×10^4	26F-62	= $6.19 \times 10^{20,521}$
26F-33	= 1.02×10^{-6}	26F-43	= -244000 = -2.44×10^5	26F-53	= 0.00385 = 3.85×10^{-3}	26F-63	= 87.4 = 8.74×10^1
26F-34	= 3.12 = 3.12×10^0	26F-44	= 7.95×10^{-6}	26F-54	= 0.228 = 2.28×10^{-1}	26F-64	= 319 = 3.19×10^2
26F-35	= 2.73×10^{-12}	26F-45	= -0.547 = -5.47×10^{-1}	26F-55	= 1.95 = 1.95×10^0	26F-65	= 3.85 = 3.85×10^0
26F-36	= 72.9 = 7.29×10^1	26F-46	= 62 integer	26F-56	= 14.0 = 1.40×10^1	26F-66	= 0.442 = 4.42×10^{-1}
26F-37	= 1.23 = 1.23×10^0	26F-47	= 72.0 = 7.20×10^1	26F-57	= 42.0 = 4.20×10^1	26F-67	= 680 = 6.80×10^2
26F-38	= 2100 = 2.1×10^3 (2SD)	26F-48	= -0.767 = -7.67×10^{-1}	26F-58	= -1350 = -1.35×10^3	26F-68	= -0.546 = -5.46×10^{-1}
26F-39	= 3.53 = 3.53×10^0	26F-49	= 5.03 = 5.03×10^0	26F-59	= 6.51 = 6.51×10^0	26F-69	= 0.572 = 5.72×10^{-1}
26F-40	= 2.22 = 2.22×10^0	26F-50	= 3610 = 3.61×10^3	26F-60	= 700 = 7.00×10^2	26F-70	= -0.0494 = -4.94×10^{-2}